

Epi and RF

Diagnosis

Definitions

Treatment

Prevention

Cases

C. diff
colonization

Detection of *C. diff* without
clinical symptoms

4-15% of healthy adults
21% of hospitalized adults
15-30% of LTC facility residents

C. diff infection (CDI)

Hospital
acquired
CDI

Community
acquired
CDI

Risk factors:

>65 yo

Healthcare exposure

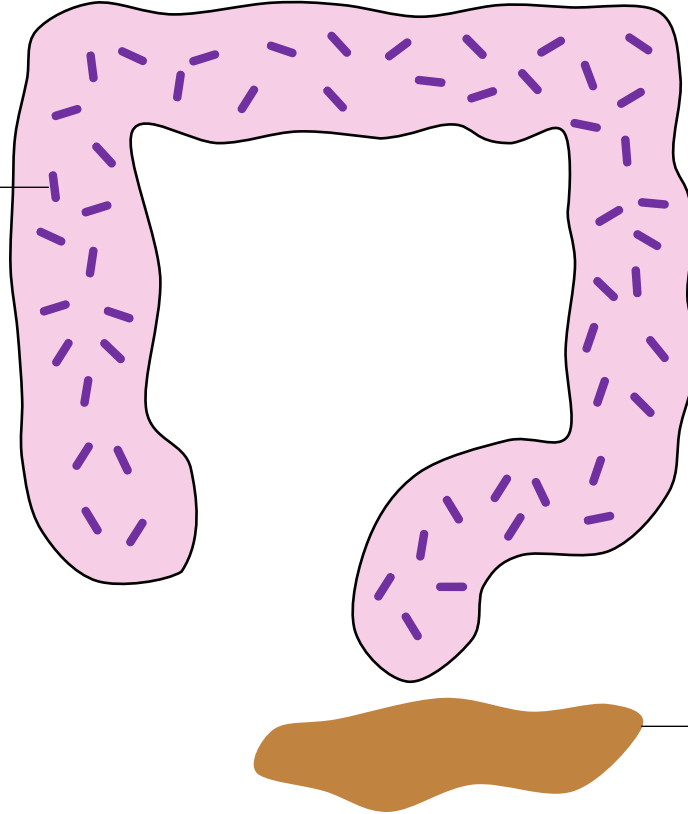
Antibiotic use

RFs for
community
acquired CDI

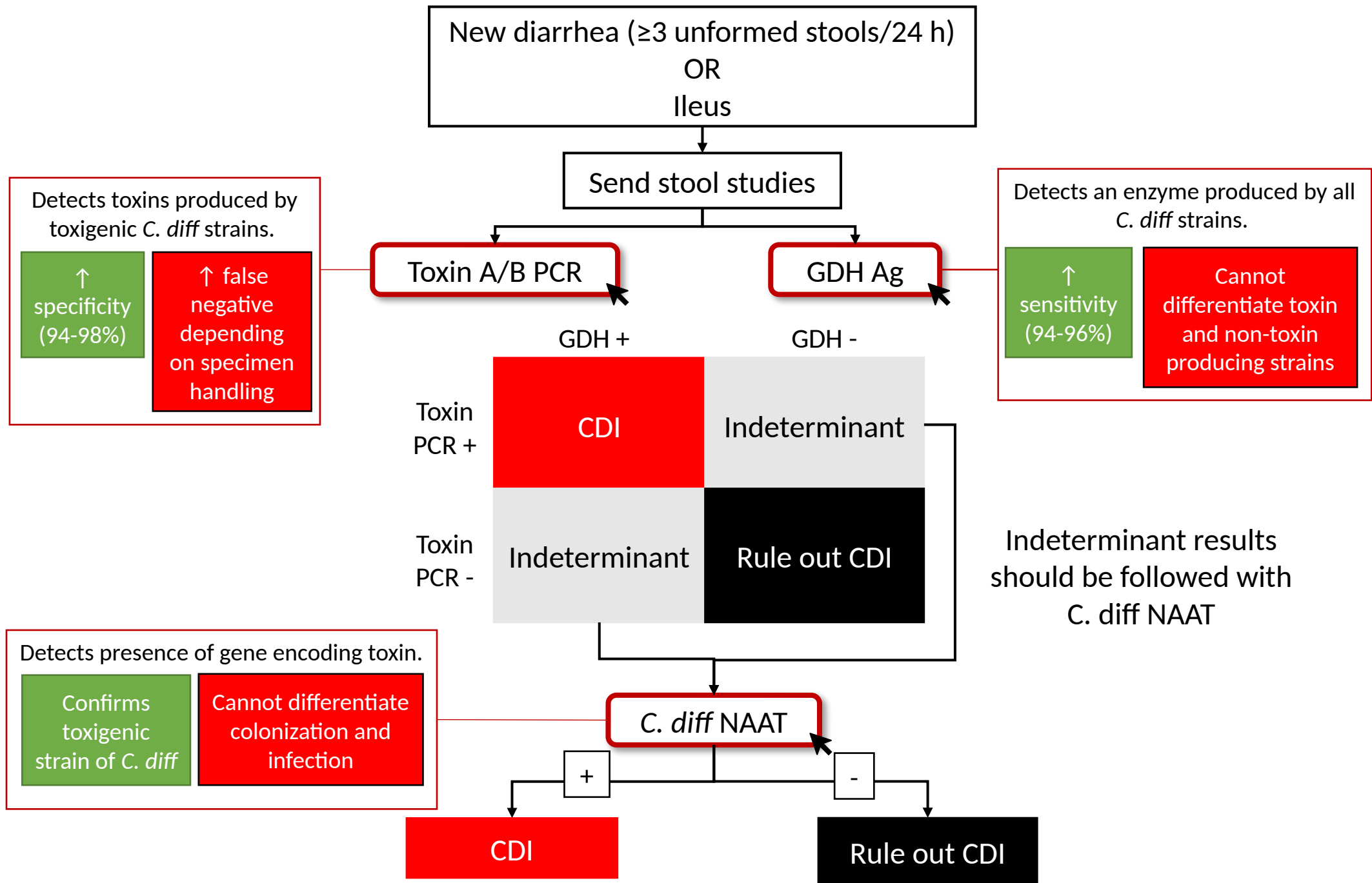
Cardiac disease

CKD

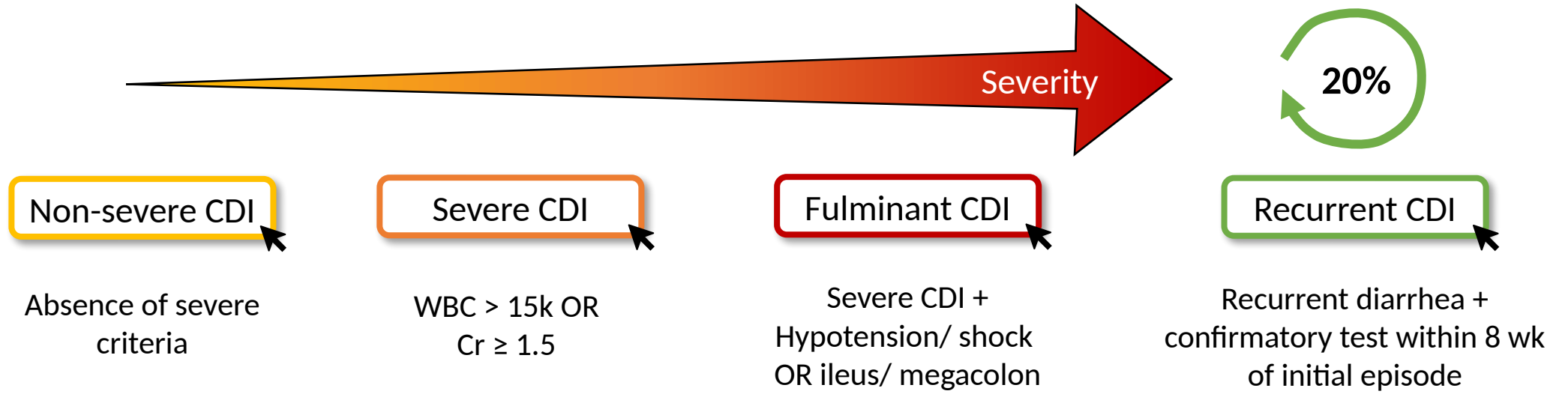
IBD

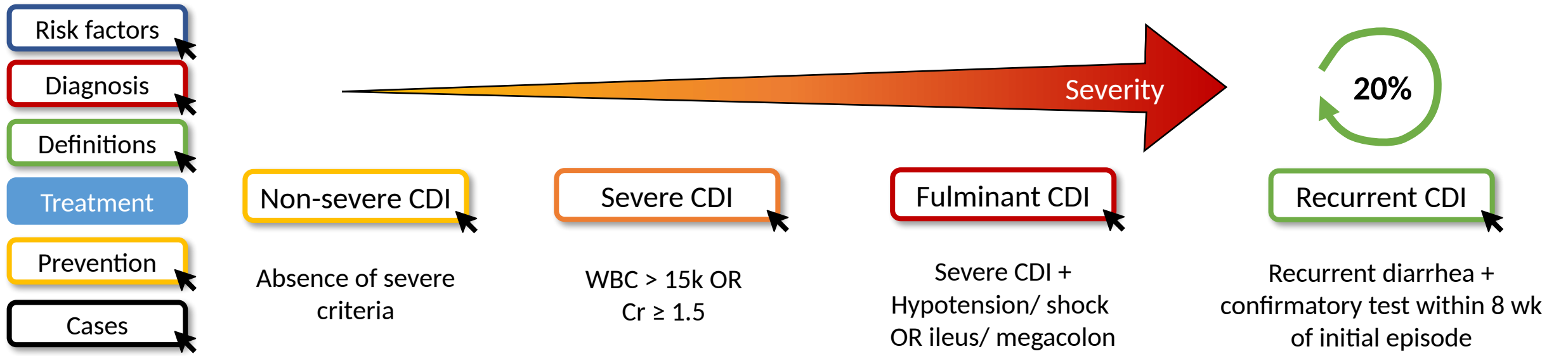


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1. Fidaxomicin 200 mg BID x 10 d
 2. PO vanc 125 mg q6h x 10 d
 PO metronidazole can be considered
 if other options are not available.



PO vanc 500 mg q6h
 ± IV metronidazole 500 mg q8h
 ± PR vanc 500 mg q6h

Surgical consultation if no
 improvement

1st recurrence:

1. Fidaxomicin x 10 d
2. Tapered/ pulse-dosed vanc
3. PO vanc x 10 d
 ± bezlotoxumab

≥ 2 recurrences:

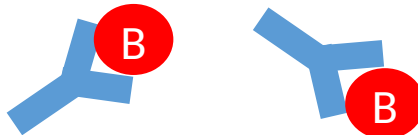
1. FMT
2. Tapered/pulsed-dose PO vanc
3. PO vanc x 10 d → rifaximin 400 mg TID x 20 d
 ± bezlotoxumab

Tapered/ pulsed vancomycin

PO vancomycin 125 mg q6h x 10-14 d
 PO vanc 125 mg BID x 7 d
 PO vanc 125 mg daily x 7 d
 PO vanc 125 mg q2-3d x 2-8 wk

Bezlotoxumab

Monoclonal antibody that binds *C. diff*
 toxin B



Risk factors

Diagnosis

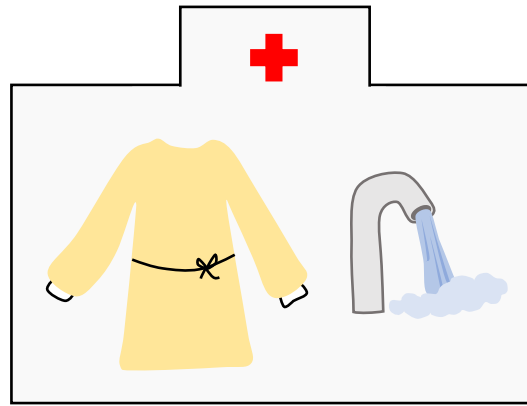
Definitions

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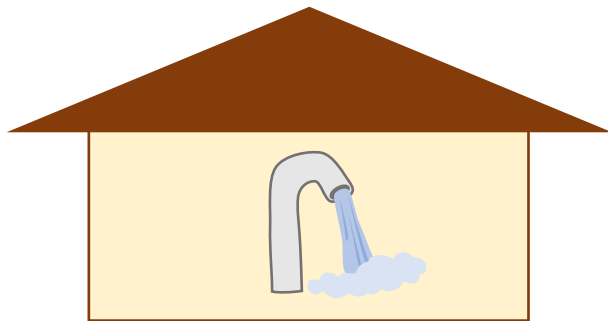
Cases

PRECAUTIONS



Contact enteric precautions (contact +
handwashing)

Until 48 h after resolution of diarrhea or until discharge



No specific precautions are needed on
discharge/ at home
(frequent handwashing is encouraged)

PREVENTION

High risk of recurrence if antibiotics are given
within 2 months of an initial episode of CDI

Probiotics

Not recommended

PO
vancomycin

125 mg daily while on systemic
antibiotics for high-risk patients

High-risk: >65 yo, immunocompromise,
hospitalized for severe CDI in past 3 mos.
Continue PO vanc for 5 days after completion
of systemic antibiotics.

Discontinuing
PPIs

Not recommended.

PPIs have been associated with higher rates
of CDI but there is insufficient evidence to
support this as a preventative measure.

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Case 1a

33 yo presents with fevers, abdominal pain and diarrhea x 4 d. They have 4-5 watery BMs per day. They were recently treated for a UTI with Bactrim.

VS: T 37.4C, HR 110, BP 100/50s

128			
	18	1.7	18

GDH Ag + Stool culture pending
C. diff toxin PCR -

Acute diarrhea. CDI is likely, but GDH and toxin PCR is indeterminant.

C. diff NAAT

C. diff NAAT came back positive.

Severe CDI. Based on a Cr > 1.5 and WBC > 15k.

PO fidaxomicin 200 mg BID or PO vancomycin 125 mg q6h x 10 days.

Case 1b

33 yo with recent diagnosis of severe CDI treated with 10 days of PO vanc, presents 5 ws later with recurrent watery diarrhea. They deny any interval antibiotics.

C. diff testing is positive.
She is otherwise afebrile, with stable vital signs. BMP and CBC are wnl.

Recurrent (non-severe) CDI. Recurrent symptoms with confirmatory testing within 8 weeks of initial CDI.

Updated guidelines from IDSA/ACG recommend **fidaxomicin** where available. An alternative would be tapering/pulse dosed **PO vancomycin**.

They are treated with a prolonged course of PO vanc. Two weeks after completion of treatment, they develop diarrhea.

Repeat *C. diff* testing to test for recurrent CDI. They are at high risk of recurrence, IBS or altered bowel patterns can also occur after CDI.

Case 2

82 yo with recurrent UTIs is admitted with fever, abdominal pain, nausea/vomiting, and constipation/obstipation x 2 d.

VS: T 38.4C, HR 110, BP 80/50s

128			
	18	1.7	18

KUB shows ileus.

Rectal swabs can be used to diagnose CDI in patients with ileus.

GDH Ag is + and toxin PCR is +.

Fulminant CDI

PO vanc 500 mg q6h (can be given via NGT) + **IV metronidazole** 500 mg q8h + **PR vanc** 500 mg q6h. Consider Surgery consult.

Two months later, they are readmitted for UTI.

Consider PO vancomycin 125 mg daily for secondary prophylaxis while on antibiotics.

Treatment Update (2023–2025)

- Fidaxomicin-preferred teaching is CURRENT: 2021 IDSA/SHEA focused update favors fidaxomicin over vancomycin for an initial episode and first recurrence; 2021 ACG also lists fidaxomicin and vancomycin as first-line (metronidazole only if neither is available / low-risk non-severe).
- BEZLOTOXUMAB DISCONTINUED: Merck discontinued bezlotoxumab (Zinplava) effective Jan 31, 2025 — it is no longer commercially available in the US. Where the slides list "± bezlotoxumab," treat that option as unavailable; rely on antibiotic strategy + microbiota-based prevention instead.
- FDA-approved live biotherapeutics for PREVENTING recurrent CDI (after antibiotics complete) now standardize what the deck calls "FMT": Rebyota (fecal microbiota, live-jslm; rectal; approved Nov 2022) and Vowst (fecal microbiota spores, live-brpk; ORAL; approved Apr 2023). Conventional FMT remains an option, often via referral.
- Unchanged: severity thresholds (WBC >15k or Cr ≥1.5), fulminant = severe + hypotension/shock or ileus/megacolon, recurrence within 8 wks, test only symptomatic patients, do not test for cure.

Take-Home Points

- Colonization is common — test ONLY symptomatic patients (≥ 3 unformed stools/24 h or ileus). Never test for cure; shedding persists after symptoms resolve.
- Diagnose with a two-step algorithm: GDH antigen + toxin A/B PCR; if discordant/indeterminate, confirm with *C. diff* NAAT.
- Severe CDI = WBC $>15k$ OR Cr ≥ 1.5 mg/dL. Fulminant = severe PLUS hypotension/shock or ileus/megacolon. Recurrence (~20%) = symptoms + positive test within 8 weeks.
- Fidaxomicin is preferred first-line for non-severe, severe, and first-recurrence CDI; PO vancomycin is an acceptable alternative. Fulminant: high-dose PO (or PR if ileus) vancomycin + IV metronidazole $\pm$ surgery.
- For recurrent CDI prevention after antibiotics: microbiota-based therapy (Rebyota, Vowst, or conventional FMT). Bezlotoxumab is no longer available (discontinued Jan 2025).

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Adapted from the TeachIM original "C. diff Infection" (published September 2021).

- Author: Yilin Zhang, MD (Assistant Professor, Department of Medicine, Valley Medical Center).
- Source: https://teachim.org/teaching_material/cdiffcolitis/
- License: Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0). Non-commercial educational use only.
- Optimization (June 2026): three surgical typo corrections — "rifaxmin" → "rifaximin" and "Bezlotuoxumab" → "Bezlotoxumab" (slide 4); pie-chart label "Hospital acquried" → "acquired" (slide 1). All original click-to-reveal animations and graphics preserved.
- Added: 2023–2025 treatment-update slide (bezlotoxumab discontinued Jan 2025; Rebyota/Vowst FDA-approved) + take-home slide. No companion handouts are published for this lesson.

Medical content should be clinician-reviewed before teaching.